

Solving Equations & Inequalities

One- & Two-Step · Multi-Step · Fractions & Distribution · Rearranging Formulas · Inequalities

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Practise solving equations, rearranging formulas and solving inequalities independently.

Show your full working in the space provided under each question. Levels: L1 fluency, L2 application, L3 thinking. Full worked solutions are in the Answer Key.

Level 1 — Fluency

1. Solve: $x + 11 = 20$. **[One-step]**

2. Solve: $2x - 6 = 10$. **[Two-step]**

3. Solve: $5x = 3x + 8$. **[Both sides]**

4. Solve: $4(x - 2) = 12$. **[Distribution]**

5. Solve $y = x + c$ for x . **[Rearrange]**

6. Solve and describe the graph: $x - 2 > 3$. **[Inequality]**

Level 2 — Application

7. Solve: $x/6 = 7$. **[One-step]**

8. Solve: $8 - 3x = 2$. **[Two-step]**

9. Solve: $9x - 4 = 5x + 12$. **[Both sides]**

10. Solve: $(x - 3)/5 = 4$. **[Fractions]**

11. Solve $P = 2l + 2w$ for l . **[Rearrange]**

12. Solve and describe the graph: $3x + 2 \leq 14$. **[Inequality]**

Level 3 — Thinking

13. The sum of two consecutive even integers is 46. Find the integers. **[Consecutive]**

14. Solve: $5(x - 1) = 3(x + 3)$. **[Both-sides distribute]**

15. Solve: $x/4 + x/2 = 9$. **[Fractions]**

16. Solve $v = u + at$ for a . **[Rearrange]**

17. Solve and describe the graph: $-3x + 1 < 10$. **[Inequality (flip)]**
18. The area of a triangle is $A = (1/2)bh$. Rearrange for h , then find h when $A = 24$ and $b = 6$. **[Formula & evaluate]**